AZURE COST SAVINGS REPORT

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Bridging Usability, Engineering & Business Strategy

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Executive Summary

This analysis re-evaluates the raw Azure cost data from four Excel workbooks (Books 1–4) covering December 21, 2025 through January 21, 2026. By comparing the first 7 days (Dec 21–27) against the last 7 days (Jan 15–21), we measure the actual cost reductions achieved through optimization efforts. This version of the report organizes the savings by the underlying optimization strategy rather than by individual Azure service, giving a clearer picture of which efforts produced the most impact.

Key Findings

METRIC	VALUE
Daily Cost Reduction (CAD)	**$1,076 / day**
First 7 Days Average (CAD)	$2,219 / day
Last 7 Days Average (CAD)	$1,143 / day
Monthly Savings (CAD)	$32,280
Annual Savings Projection (CAD)	**$392,740**

Savings by Optimization Strategy

The cost reductions achieved during this period can be attributed to five distinct optimization strategies. The following table summarizes the estimated daily savings contribution from each strategy.

OPTIMIZATION STRATEGY	EST. SAVINGS (CAD/DAY)	EST. SAVINGS (CAD/MO)	SHARE OF TOTAL
1. Environment Decommissioning	**~$483**	~$14,490	46.0%
2. Orphaned Resource Cleanup	**~$146**	~$4,380	13.9%
3. Customer Migration & ASP Processing Tier Downgrades	**~$120**	~$3,600	11.4%
4. Targeted Scope Optimization	**~$95**	~$2,850	9.0%
5. KEDA Autoscaling	**~$38**	~$1,140	3.6%
6. Stop Runaway Process	**~$54**	~$1,620	5.1%
7. Temporary Disabling of New US AKS Cluster	**~$116**	~$3,480	11.0%
TOTAL	~$1,050	~$31,500	100%

Note: Strategy attribution is estimated. Some resource groups were affected by multiple strategies simultaneously.

1. Environment Decommissioning

Estimated savings: ~$483 CAD/day (~46% of total reduction)

The single largest driver of cost reduction was the decommissioning process for soon to be unused environments and clusters that were no longer going to serve production workloads. This was a coordinated effort that removed infrastructure across multiple Azure services over a period starting on December 21, 2005, producing cascading savings as each dependent resource was eliminated.

The ca-legacy-aks resource group was fully decommissioned, removing all associated VMs, managed disks, storage accounts, and networking resources. The us-legacy-aks resource group was reduced by 98.5%, retaining only minimal residual resources. The legacy us-stg-aks staging environment was fully removed.

AFFECTED SERVICE	FIRST 7 DAYS	LAST 7 DAYS	SAVINGS / DAY
Virtual Machines	$468	$106	**-$362**
Storage (orphaned disks/accounts)	$158	$22	**-$136**
Container Registry	$43	$9	**-$34**
Virtual Machine Licenses	$45	$22	**-$23**
Redis Cache	$41	$23	**-$18**
Backup (reduced scope)	$110	$96	**-$14**
Application Gateway	$61	$51	**-$10**
TOTAL / DAY	$926	$329	**-$597**

1.1 Virtual Machines

The decommissioning of AKS clusters produced the largest savings. The ca-legacy-aks resource group went from $197 CAD/day to $0 (complete removal).

Resource Group Breakdown

RESOURCE GROUP	FIRST 7 DAYS	LAST 7 DAYS	SAVINGS/DAY
ca-legacy-aks	$197	$0.00	**-$197**
us-v3-aks	$118	$2	**-$116**
ca-v3-aks	$142	$104	**-$38**
us-stg-aks	$10	$0.00	**-$10**
Other	$33	$31	**-$2**

VM SKU Breakdown

METER	FIRST 7 DAYS	LAST 7 DAYS	SAVINGS/DAY
D8as v5	$169	$47	**-$122**
B4as v2	$130	$38	**-$92**
D8 v3/D8s v3	$79	$0.49	**-$79**
B4ms	$34	$1	**-$32**
B2s	$25	$8	**-$17**
B4als v2	$20	$9	**-$11**
E2 v3/E2s v3	$4	$0.00	**-$4**
B2ms	$7	$3	**-$3**

1.2 Virtual Machine Licenses

This was a direct byproduct of the Virtual Machines decommissioning. As VMs were removed, their associated OS and software license charges dropped proportionally. No separate license-specific actions were required.

1.3 Storage

Orphaned managed disks and storage accounts left behind by the retired VM environments were identified and cleaned up, eliminating ongoing charges for storage that was no longer referenced by any active workload.

The opportunity was taken to test the Elasticsearch Node and Disk Reduction procedure (detailed in the Elasticsearch Reduction Runbook) on an Elasticsearch cluster that was about to be decommissioned, validating that the process worked correctly before applying it to production clusters in the future.

Resource Group Breakdown

RESOURCE GROUP	FIRST 7 DAYS	LAST 7 DAYS	SAVINGS/DAY
ca-prod-elastic	$53	$9	**-$44**
ca-qa-aks	$48	$4	**-$44**
us-stg-aks	$14	$0.11	**-$14**
ca-legacy-aks	$14	$1	**-$13**
us-prod-elastic	$9	$2	**-$7**
us-stg-elastic	$10	$3	**-$7**
us-aks-legacy	$10	$3	**-$6**

Storage Meter Breakdown

METER	FIRST 7 DAYS	LAST 7 DAYS	SAVINGS/DAY
P10 LRS Disk	$88	$29	**-$59**
Cool RA-GRS Data Stored	$56	$3	**-$53**
Hot GRS Write Operations	$60	$18	**-$42**
P15 LRS Disk	$29	$7	**-$23**
Premium LRS Provisioned Capacity	$19	$8	**-$10**
GRS List and Create Container Operations	$13	$8	**-$5**
Class 2 Operations	$8	$3	**-$5**
P20 LRS Disk	$3	$0.00	**-$3**
LRS Data Stored	$7	$4	**-$3**
LRS Write Operations	$4	$2	**-$2**

1.4 Container Registry

The decommissioning of old infrastructure QA environments included their associated container registries. The us-qa-prodops registry dropped 89% as images and repositories from the retired environments were cleaned up.

Resource Group Breakdown

RESOURCE GROUP	FIRST 7 DAYS	LAST 7 DAYS	SAVINGS/DAY
us-qa-prodops	$35	$4	**-$31**
us-stg-prodops	$2	$0.22	**-$2**

1.5 Redis Cache

Redis Cache instances associated with the decommissioned environments were removed. Several QA system caches (ca-q2-sys, us-q1-sys, us-q2-sys) were fully eliminated, and the main API cache (ca-legacy-api) dropped approximately 49%.

Resource Group Breakdown

RESOURCE GROUP	FIRST 7 DAYS	LAST 7 DAYS	SAVINGS/DAY
ca-legacy-api	$23	$12	**-$11**
us-qa-shared	$5	$2	**-$4**
ca-q2-sys	$0.74	$0.00	**-$0.74**
us-q1-sys	$0.74	$0.00	**-$0.74**
us-q2-sys	$0.74	$0.00	**-$0.74**
ca-q1-sys	$0.74	$0.20	**-$0.54**

1.6 Backup

The backup scope was reduced following the Elasticsearch node reduction — fewer nodes and fewer disks meant less data to back up. Backup retention policies were also reviewed and adjusted to reduce the volume of historical snapshots being retained.

Resource Group Breakdown

RESOURCE GROUP	FIRST 7 DAYS	LAST 7 DAYS	SAVINGS/DAY
ca-prod-backup	$100	$87	**-$13**
us-prod-backup	$9	$8	**-$1**

1.7 Application Gateway

This was a byproduct of the QA environment decommissioning. The QA Elasticsearch Application Gateway (qa-us-elastic) dropped 72% as the associated environment was retired.

Resource Group Breakdown

RESOURCE GROUP	FIRST 7 DAYS	LAST 7 DAYS	SAVINGS/DAY
us-qa-elastic	$14	$4	**-$10**

2. Orphaned Resource Cleanup

Estimated savings: ~$146 CAD/day (~13.9% of total reduction)

Beyond full environment decommissioning, a targeted effort was made to identify and remove orphaned resources — individual containers, databases, and test tenants that were no longer actively used but continued to incur costs.

2.1 Azure Cosmos DB

Orphaned Cosmos DB containers that were no longer actively used were identified and removed. Each container carried provisioned RU/s regardless of whether it was receiving traffic, so eliminating unused containers directly reduced the total provisioned throughput required.

Resource Group Breakdown

RESOURCE GROUP	FIRST 7 DAYS	LAST 7 DAYS	SAVINGS/DAY
us-prod-api	$46	$2	**-$43**
ca-prod-api	$41	$5	**-$36**
ca-q1-sys	$16	$2	**-$14**
ca-qa-sys	$13	$4	**-$8**
us-prod-auth	$3	$0.07	**-$3**
ca-dev04-sys	$3	$0.00	**-$3**
ca-q2-sys	$3	$0.00	**-$3**
us-stg-auth	$1	$0.03	**-$1**

2.2 SQL Database — Redundant Test Tenants

A significant number of redundant test tenants in QA and staging environments were identified and removed. This reduced the database footprint in us-qa-shared by 52% and ca-legacy-api by 87%.

Resource Group Breakdown

RESOURCE GROUP	FIRST 7 DAYS	LAST 7 DAYS	SAVINGS/DAY
ca-legacy-api	$11	$1	**-$9**
us-qa-shared	$14	$6	**-$7**
ca-q2-sys	$4	$0.00	**-$4**
us-q2-sys	$3	$0.00	**-$3**
us-q1-sys	$3	$0.00	**-$3**
ca-q1-sys	$4	$1	**-$3**
ca-prod-sys	$23	$22	**-$0.70**
team-unittest	$3	$2	**-$0.68**

3. Customer Migration & Tier Downgrades

Estimated savings: ~$120 CAD/day (~11.4% of total reduction)

As customers were migrated from the old infrastructure to the new platform, the load on the legacy App Service plans dropped substantially. This enabled tier downgrades across multiple App Service plans without impacting service quality.

The resources that were using Azure P1v3 App Service Plans was scaled down significantly, and several resource groups saw 88–90% reductions as tenants moved off the legacy services. The main app service plan (prod-ca-appserviceplan) was reduced by 59%.

Resource Group Breakdown

RESOURCE GROUP	FIRST 7 DAYS	LAST 7 DAYS	SAVINGS/DAY
ca-legacy-appserviceplan	$75	$31	**-$44**
ca-legacy-api-index	$29	$3	**-$26**
ca-legacy-api-ext	$15	$2	**-$13**
ca-legacy-previewer	$11	$1	**-$10**
ca-legacy-exchange	$9	$1	**-$8**
ca-legacy-onprem	$9	$1	**-$8**
us-legacy-contentprocessing	$11	$6	**-$6**
ca-legacy-contentprocessing	$6	$0.64	**-$5**

4. Targeted Scope Optimization

Estimated savings: ~$95 CAD/day (~9% of total reduction)

These savings came from reviewing the configuration and scope of monitoring and security services, and adjusting them to match actual requirements rather than applying blanket coverage.

4.1 Microsoft Defender for Cloud

An audit was conducted to determine which storage accounts genuinely required Defender for Storage features. Coverage was selectively enabled only for accounts that warranted the protection.

Resource Group Breakdown

RESOURCE GROUP	FIRST 7 DAYS	LAST 7 DAYS	SAVINGS/DAY
us-stg-prodops	$15	$0.00	**-$15**
us-qa-shared	$10	$0.26	**-$10**
ca-legacy-aks	$9	$0.73	**-$8**
ca-sys	$31	$23	**-$7**
ca-legacy-appserviceplan	$7	$5	**-$2**
us-stg-elastic	$2	$0.00	**-$2**
ca-legacy-api-index	$2	$0.26	**-$2**
ca-prod-previewer	$3	$0.91	**-$2**

4.2 Log Analytics

Data ingestion volume and sampling rates were reduced to lower the amount of telemetry flowing into Log Analytics workspaces. The primary workspace (ca-prod-ops) saw a 67% reduction in daily ingestion costs.

Resource Group Breakdown

RESOURCE GROUP	FIRST 7 DAYS	LAST 7 DAYS	SAVINGS/DAY
ca-prod-ops	$28	$9	**-$19**
ca-q1-ops	$6	$0.00	**-$6**
ca-q2-ops	$5	$0.00	**-$5**
ca-prod-prodops	$4	$0.23	**-$4**
default-rg-westus	$4	$0.00	**-$4**
us-sys	$3	$0.06	**-$3**
us-q1-ops	$3	$0.00	**-$3**
ca-qa-ops	$4	$2	**-$2**

5. KEDA Autoscaling

Estimated savings: ~$38 CAD/day (~3.6% of total reduction)

In parallel with the environment decommissioning, KEDA-based (Kubernetes Event-Driven Autoscaling) pod scaling rules were introduced across the remaining AKS clusters. These rules use Azure queue size and Kafka topic lag as scaling triggers, allowing compute to scale down aggressively during periods of low demand rather than running at fixed capacity.

This is most visible in the ca-v3-aks resource group, which dropped from $142 CAD/day to $104 CAD/day (27% reduction) despite remaining a fully active production cluster. The KEDA autoscaling rules provide ongoing, compounding savings as they adapt to actual workload demand in real time.

Resource Group Breakdown

RESOURCE GROUP	FIRST 7 DAYS	LAST 7 DAYS	SAVINGS/DAY
ca-v3-aks	$142	$104	**-$38**

6. Runaway Process for Specific Customer

Estimated savings: ~$54 CAD/day (~5.1% of total reduction)

We uncovered a runaway process within ca-prod-tenant-xxxxx that was resolved.

Resource Group Breakdown

RESOURCE GROUP	FIRST 7 DAYS	LAST 7 DAYS	SAVINGS/DAY
ca-prod-tenant-xxxxx	$65	$11	**-$54**

7. Temporary Disabling of New US AKS Cluster

Estimated savings: ~$116 CAD/day (~11% of total reduction)

The us-v3-aks resource group dropped from $118 CAD/day to $2 CAD/day (98.5% reduction) as we temporarily disabled the new Kubernetes cluster until we were ready for tenant migrations from legacy to v3 infrastructure.

Additional cost savings with decommissioning the legacy US environments will offset the costs of the new v3 infrastructure being restarted.

Resource Group Breakdown

RESOURCE GROUP	FIRST 7 DAYS	LAST 7 DAYS	SAVINGS/DAY
us-v3-aks	$118	$2	**-$116**

8. Areas of Concern – Cost Increases

SERVICE	FIRST 7 DAYS	LAST 7 DAYS	DAILY INCREASE	% CHANGE
Key Vault	$11	$20	**+$9**	+86.3%

Key Vault (+$9 CAD/day): Premium HSM-protected key usage approximately doubled during the period. This requires investigation.

Foundry Tools (+$0.25 CAD/day): New usage appeared during the period. While the absolute cost is negligible, the percentage increase (357%) warrants review.

Summary

METRIC	VALUE (CAD)
Daily Cost Reduction Achieved	**$1,076/day**
Percentage Reduction	**48.5%**
Monthly Savings (30 days)	**$32,280**
Annual Savings Projection	**$392,740**

The 48.5% reduction in daily Azure spending was achieved through five complementary strategies: Environment Decommissioning (~46% of savings), Orphaned Resource Cleanup (~13.9%), Customer Migration & ASP Processing Tier Downgrades (~11.4%), Targeted Scope Optimization (~9%), KEDA Autoscaling (~3.6%), identifying and stopping a runaway process (~5.1%), and temporarily disabling an unused environment (~11%). Only Key Vault (+$9/day) and Foundry Tools (+$0.25/day) showed cost increases.

The annualized savings projection stands at approximately **$392,740 CAD**.

Appendix A: Service-Level Summary

SERVICE	FIRST 7 DAYS	LAST 7 DAYS	DAILY SAVINGS	REDUCTION
Virtual Machines	$505	$142	**-$363**	-71.9%
Storage	$402	$194	**-$208**	-51.8%
Azure App Service	$311	$171	**-$140**	-45.1%
Azure Cosmos DB	$223	$90	**-$133**	-59.5%
Log Analytics	$101	$54	**-$47**	-46.6%
Microsoft Defender for Cloud	$168	$122	**-$47**	-27.7%
Container Registry	$43	$9	**-$34**	-78.7%
SQL Database	$98	$66	**-$32**	-32.5%
Virtual Machines Licenses	$45	$22	**-$24**	-52.3%
Redis Cache	$41	$23	**-$18**	-43.1%
Backup	$110	$96	**-$14**	-12.7%
Application Gateway	$61	$50	**-$12**	-19.0%
Azure DevOps	$47	$41	**-$6**	-13.0%
Load Balancer	$24	$19	**-$5**	-21.4%
Event Hubs	$3	$2	**-$1**	-35.4%
Virtual Network	$8	$8	**-$0.61**	-7.3%
TOTAL / DAY			**-$1,084**	

Appendix B: Daily Cost Data (CAD)

DATE	DAILY COST (CAD)	DAILY COST (USD)	VS. BASELINE
2025-12-21	$2,314	$1,640	+$95
2025-12-22	$2,344	$1,662	+$126
2025-12-23	$2,216	$1,571	-$2
2025-12-24	$2,210	$1,567	-$8
2025-12-25	$2,134	$1,513	-$84
2025-12-26	$2,159	$1,530	-$60
2025-12-27	$2,153	$1,526	-$66
2025-12-28	$2,158	$1,530	-$60
2025-12-29	$2,220	$1,574	+$2
2025-12-30	$2,199	$1,559	-$19
2025-12-31	$2,098	$1,487	-$120
2026-01-01	$2,040	$1,492	-$179
2026-01-02	$1,980	$1,448	-$239
2026-01-03	$1,901	$1,391	-$317
2026-01-04	$1,875	$1,372	-$344
2026-01-05	$2,068	$1,513	-$151
2026-01-06	$2,057	$1,505	-$161
2026-01-07	$1,944	$1,422	-$274
2026-01-08	$1,978	$1,447	-$240
2026-01-09	$1,968	$1,440	-$250
2026-01-10	$1,831	$1,340	-$387
2026-01-11	$1,534	$1,122	-$685
2026-01-12	$1,439	$1,053	-$780
2026-01-13	$1,464	$1,071	-$754

2026-01-14	$1,473	$1,077	-$746
2026-01-15	$1,419	$1,038	-$800
2026-01-16	$1,232	$901	-$987
2026-01-17	$1,098	$803	-$1,121
2026-01-18	$1,067	$780	-$1,152
2026-01-19	$1,103	$807	-$1,116
2026-01-20	$1,041	$762	-$1,177
2026-01-21	$1,042	$762	-$1,177